

# LOUIS PASTEUR

French biologist and chemist  
1822–1895

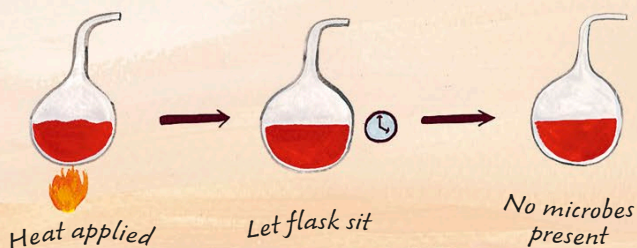
Scientists wanted to use Edward's ideas to protect people against other diseases, but it wasn't that easy. Other diseases didn't have milder forms like smallpox, so it would take a new technique to create vaccinations for them. Louis Pasteur figured out how to do just that.

Louis' work began with his discovery that microbes (tiny living things) were responsible for spoiling wine. He determined that by gently heating the wine, he would kill any harmful microbes in it. This process is called "pasteurization" and is still used to help milk, fruit juice, and other foods last longer on supermarket shelves.

## Microbe magician

After Louis found that heating deadly microbes made them weaker, he injected them into animals. The microbes weren't strong enough to make the animals sick, but their immune systems learned how to recognize and fight the invaders. If one of the animals came across the same microbes in the future, its immune system destroyed them. Although this new technique had nothing to do with cowpox, Louis kept the name vaccination in honor of Edward.

## Louis' pasteurization process



## Vaccinations

Today, vaccinations are still made using dead or weakened microbes. Dozens of diseases that were once common and deadly can now be controlled and prevented, such as tetanus, measles, and polio. This saves millions of lives every year.

